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Heavy metal removal by hydrogel synthesized from rice bran/acrylic acid/sodium alginate

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This study reports the synthesis of a novel hydrogel adsorbent prepared from rice bran (RB), acrylic acid (AA), and sodium alginate (SA) for the removal of Cu^{2+} and Ni^{2+} from aqueous solutions. RB, an abundant agricultural by-product rich in cellulose and functional groups, was combined with SA and polymerized with AA to produce a cost-effective, eco-friendly, and reusable adsorbent. Adsorption performance was systematically evaluated under varying pH values, temperatures, metal ion concentrations, and adsorbent dosages. The optimum conditions for Cu^{2+} removal were pH 5.5, 30 °C, and an adsorbent dosage of 1 pcs, yielding a maximum adsorption capacity of 26.5 mg g^{-1} with 88% removal efficiency. For Ni^{2+}, the best performance was achieved at pH 8 and 30 °C, with a capacity of 41.81 mg g^{-1} and 62% removal efficiency. Isotherm analysis showed that adsorption followed the Langmuir model, while kinetics were best described by the pseudo-second-order model. The adsorbent retained over 85% of its initial capacity after five adsorption–desorption cycles, confirming good reusability.

Keywords Adsorption, RB, Heavy metals, Hydrogel

The World Water Council has estimated that by 2030, as many as 3.9 billion individuals will be living in water scarce areas[1]. One of the primary causes is water pollution resulting from the discharge of industrial effluent. Organic compounds and heavy metal ions in industrial wastewater tend to accumulate in the human body through the food chain, posing a serious threat to both the environment and human health.

Effluent associated with the fabrication of circuit boards typically contains high concentrations of metals and/or metalloids with atomic densities exceeding 4 $g{\cdot}cm^{-3}$[2], such as cobalt, nickel, tin, arsenic, chromium, zinc, mercury, lead, cadmium, copper, boron, and titanium[3]. The most common heavy metal ions in this effluent are Cu^{2+} and Ni^{2+}[4], with emission concentrations ranging from 10 to 150,000 $mg{\cdot}L^{-1}$. Although Cu and Ni are essential to human health, excess Cu concentrations can have adverse health effects (e.g., nausea, vomiting and kidney failure) and long-term exposure to Ni can cause chronic bronchitis and lung disease[5,6]. Note that this type of pollution is particularly severe in Taiwan—a major producer of circuit boards within a relatively small geographic area.

Numerous methods have been developed for treating heavy metal wastewater, include ion exchange[7], chemical precipitation[8,9], flocculation[10], coagulation[11], membrane separation[12], reverse osmosis[13], and adsorption[14,15,16]. Among these, adsorption is the most widely used method due to its simplicity, efficiency, low-cost, ease of manufacture, and minimal secondary pollution[17–19]. Many research teams are developing low-cost adsorbents from agricultural and household waste, sludge generated in industrial processes, marine materials, or ore[20–22]. Agricultural waste—such as agave, rice bran, bagasse, barley straw and fruit peel—has attracted particular attention due to its availability and affordability[23,24]. This is because such wastes are abundantly generated as by-products of agricultural activities, are renewable in nature. Moreover, they often contain cellulose, hemicellulose, lignin, and various functional groups capable of interacting with heavy metal ions, making them promising candidates for adsorption. However, its relatively poor adsorption performance has thus far limited its practical applicability in heavy metal removal[25].

RB is a low-cost by-product of white rice production with an estimated annual production of 100,000 tons in Taiwan. Several researchers have explored the use of RB as an adsorbent, focusing on functional group modification to enhance adsorption efficiency[26]. In the current study, RB was combined with SA to form an

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efficient adsorbent using a straightforward method that is economical, highly efficient, and environmentally friendly.

Materials and methods

Reagents and materials

N, N'-Methylenebisacrylamide ($C_7H_{10}N_2O_2$), hydrochloric acid (HCl) and nickel(II) chloride ($NiCl_2$) were purchased from Acros Organics (Geel, Belgium). SA ($(C_6H_8O_6)_n$) was purchased from Sigma–Aldrich Co. Ltd. (USA). Sodium hydroxide (NaOH) was purchased from Fisher chemical company. AA ($C_3H_4O_2$), copper(II) chloride ($CuCl_2$) and ammonium persulfate ($(NH_4)_2S_2O_8$) were purchased from Alfa Aesar chemicals Co., Ltd. All materials were prepared and applied without further purification. Water was deionized using a Milli-Q water purification system (Millipore Corp.).

Adsorbent Preparation

RB (Tainan No. 11 rice) was purchased from a commercial vendor in Taichung, Taiwan. The RB was first sun-dried for two to three days and then passed through a 100-mesh sieve (< 0.149 m·m^{-1}).

As shown in Fig. 1, the reaction solution was prepared by heating 60 mL of deionized water to 70 ° C in a square mold (8 × 8 × 1.5 cm), to which was added 8 ml of AA containing 300 mg of a cross-linking agent (N, N' – Methylenebisacrylamide; MBA) under stirring for 20 min. To this solution was added 500 mg of RB with 1200 mg of SA, and 240 mg of ammonium persulfate (APS), followed by vigorous stirring for 2 h.

The resulting hydrogel was molded into 1 cm x 1 cm cubic blocks and dried in an oven at 50 ° C over a period of 24 h. Hereafter, the RB/AA/SA Hydrogel samples are referred to as RASH.

Adsorbent characterization

The micro-structure and morphology of as-prepared RASH samples were characterized using a Scanning Electron Microscope (SEM) (JEOL JSM-7800 F, Japan). The elemental composition of as-prepared samples was analyzed using an Energy Dispersive X-ray Spectroscopy (EDS).

The chemical composition of the samples was confirmed using an X-ray photoelectron Spectroscope (ULVAC-PHI 5000 VersaProbe Scanning ESCA Microprobe) equipped with Al Kα radiation. Fourier-transform Infrared (FTIR) spectra were acquired over a wavelength range of 500–4000 cm^{-1}. The thermal decomposition of as-prepared samples was assessed using thermogravimetric analysis. The swelling of samples was calculated as follows:

$$\text{Swelling percent} = (W_t - W_d)/W_d \times 100\%)$$

where W_t indicates the weight of the swollen sample, and W_d indicates the weight of the dried sample.

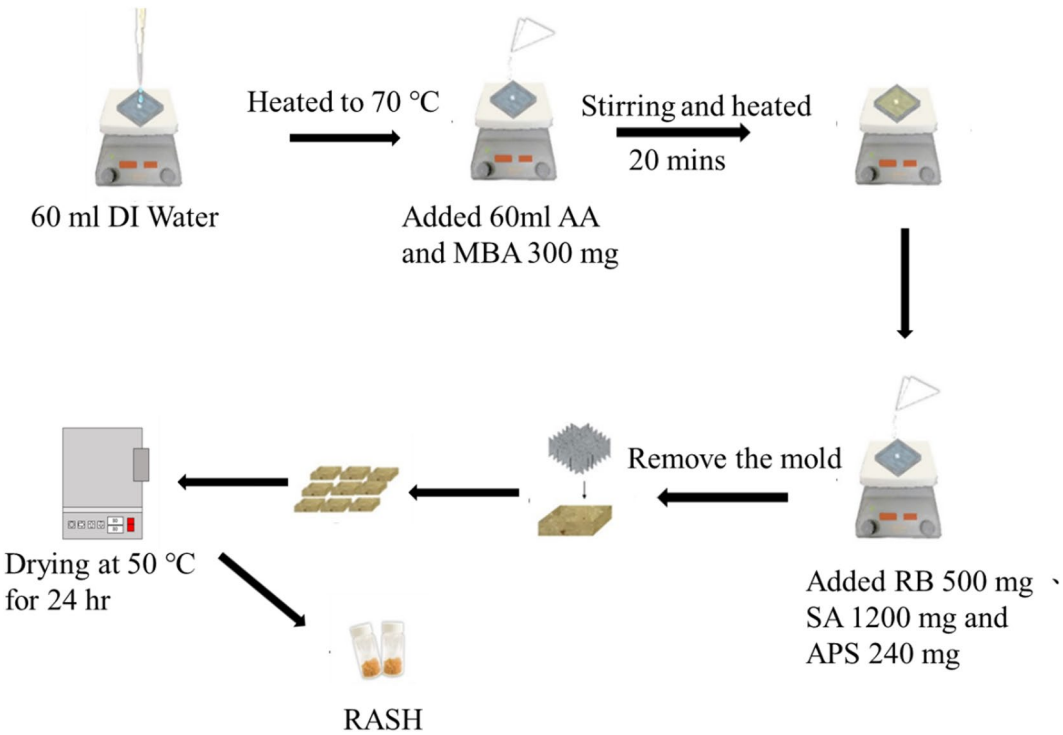

Fig. 1. Preparation flowchart of RASH adsorbent.

Batch adsorption experiments

Batch experiments were conducted to assess the individual and simultaneous adsorption of Cu(II) and Ni(II) onto as-prepared adsorbent in aqueous solutions. Stock solutions containing unitary (Cu, Ni) and binary (Cu–Ni) metal contaminants were prepared by dissolving $NiCl_2$ and $CuCl_2$ into deionized water at a concentration of 1000 ppm. Working solutions of various concentrations were subsequently obtained by diluting the stock solutions with deionized water.

Experiments were performed in a temperature-controlled water bath shaker at a mixing speed of 100 rpm for set durations. Adsorption performance was assessed under various adsorption conditions by varying key parameters, including contact time, initial concentration, pH, and temperature. The pH values of the contaminant solutions were controlled through the drop-wise addition of HCl/NaOH solutions, as follows: Cu (pH 2–6), Ni (pH 2–8).

Upon the completion of the adsorption process, the adsorbent was filtered and washed using deionized water. The initial and final metal concentrations in the solutions were measured using an ultraviolet/visible light spectrometer.

The regeneration ability of the adsorbent was also investigated by performing five Cu(II) and Ni(II) adsorption/desorption (A/D) cycles using 0.1 M AA aqueous solution. Metal ion adsorption and adsorption capacity were respectively calculated using the following equations:

$$\mathrm{R} = \frac{C_0 - C_e}{C_0} \times 100\% \tag{1}$$

$$Q_e = \frac{V(C_0 - C_e)}{m} \tag{2}$$

where Q_e refers to the amount ($mg{\cdot}g^{-1}$) of metal ions adsorbed by the adsorbent, C_0 and C_e respectively indicate the metal concentrations ($mg{\cdot}L^{-1}$) in the solution before and after adsorption, V is the volume (L) of the solution, and m is the mass (g) of adsorbent used in the experiments.

The adsorption process was analyzed in terms of its similarity to pseudo-first-order and pseudo-second-order rate expressions, which were used to describe the adsorption kinetics:

Pseudo first-order rate expression:

$$\ln\left[Q_0 - Q_t\right] = lnQ_0 - K_1 t \tag{3}$$

Pseudo second-order rate expression:

$$\frac{t}{Q_t} = \frac{1}{K_2 Q_0^2} + \frac{1}{Q_0} t \tag{4}$$

where Q_0 is the equilibrium adsorption capacity ($mg{\cdot}g^{-1}$), Q_t is the adsorption capacity over time, k_1 is the pseudo-first-order rate constant (min^{-1}), k_2 is the pseudo-second-order rate constant (g (mg min)$^{-1}$).

The Langmuir, Freundlich, and Dubinin–Radushkevich (D–R) models were selected to describe the adsorption isotherm more scientifically. These models represent the adsorption modes and interactions between adsorbed molecules or ions. The Langmuir isotherm model assumes that adsorption involves a saturated monolayer of solute molecules on a homogeneous surface, and that adsorption energy remains constant without lateral interactions or the migration of adsorbate particles. The model is typically represented using the following equation:

$$Q_e = \frac{Q_0 K_L C_e}{1 + K_L C_e} \tag{5}$$

where Q_e refers to the equilibrium adsorption capacity of the adsorbate on the adsorbent ($mg{\cdot}g^{-1}$), C_e indicates the equilibrium concentration of the adsorbate in the contaminant solution ($mg{\cdot}L^{-1}$), Q_0 represents the maximum monolayer capacity of adsorbent ($mg{\cdot}g^{-1}$), and K_L is the Langmuir adsorption constant ($L{\cdot}mg^{-1}$), related to the free energy of adsorption.

The Freundlich model assumes a heterogenous energetic distribution of sites due to the diverse nature of adsorption sites or the varied forms of adsorbed metal ions, whether free or hydrolyzed. Unlike the Langmuir model, the Freundlich model is an empirical isotherm describing multilayer adsorption on a heterogeneous surface. It accounts for the non-uniform distribution of adsorption energies through the heterogeneity factor (n^{-1}). It describes reversible adsorption and is not restricted to the formation of a monolayer, which makes it useful for analyzing adsorption data. The model is expressed as follows:

$$Q_e = K_F C_e^{\frac{1}{n}} \tag{6}$$

where K_F ($mg{\cdot}g^{-1}$ $(L{\cdot}mg^{-1})^{1/n}$) and n^{-1} represent the Freundlich constants corresponding to adsorption capacity and adsorption intensity, respectively.

The Dubinin–Radushkevich equation is widely used to describe adsorption in microporous materials, as follows:

$$\varepsilon = \mathrm{RTln}\left(1 + \frac{1}{C_e}\right) \tag{7}$$

where C_e represents the equilibrium concentration in solution ($mg \cdot L^{-1}$).

Kinetic data can also be analyzed using the intraparticle diffusion kinetics model

$$q_t = K_{id} t^{0.5} + C \tag{8}$$

where k_i ($mg \cdot g^{-1} \cdot h^{-1/2}$) is the intraparticle diffusion rate constant and C ($mg \cdot g^{-1}$) is a constant.

Results and discussion

Figure 2a presents SEM images of as-prepared adsorbent illustrating the morphology and microstructure. In its raw form, the RB presented a distinct shape with a rough surface[27]. When combined with AA to form RB/AA, the surface of the bran became smoother (Fig. 2b) with a hydrogel coating that hid individual particles and pores. When combined with AA and alginate to form RASH, the surface tightened and wrinkled (Fig. 2c). The appearance of a few surface pores was likely caused by the loss of water molecules during drying. The porous structure with surface folds increased the surface area, resulting in a larger number of adsorption sites, thereby enhancing adsorption performance.

As shown in Fig. 3, EDS analysis was used to identify the elements in RASH samples. The as-prepared adsorbent presented mainly carbon and oxygen atoms (Fig. 3a), however, peaks indicative of Cu and Ni appeared after the adsorption process (Fig. 3b and c, and 3d).

Fourier-transform infrared spectroscopy were used to study functional groups on the as-prepared adsorbents. As shown in Fig. 4a, the RB exhibited a wide absorption peak at 3200–3600 cm^{-1}, indicating the presence of O-H and N-H groups from glutin, ferulic acid, and vitamins in the RB[28]. Peaks detected at 2933 cm^{-1} and 2855 cm^{-1} corresponded to the stretching vibration of C-H, and an obvious peak at 1712 cm^{-1} corresponded to C = O groups from glutaminol and ferulic acid. A peak detected at 1595 cm^{-1} corresponded to the OH stretching vibration of silanol groups[29].

Absorption bands at 1350–1480 cm^{-1} indicated the stretching vibrations of O–H, while a peak at 1014 cm^{-1} corresponded to the stretching vibrations of C-O. In RB/AA, a peak at 1718 cm^{-1} was associated with overlapping stretching vibrations of carboxylamine and NH bending vibrations of an amide group. The peaks

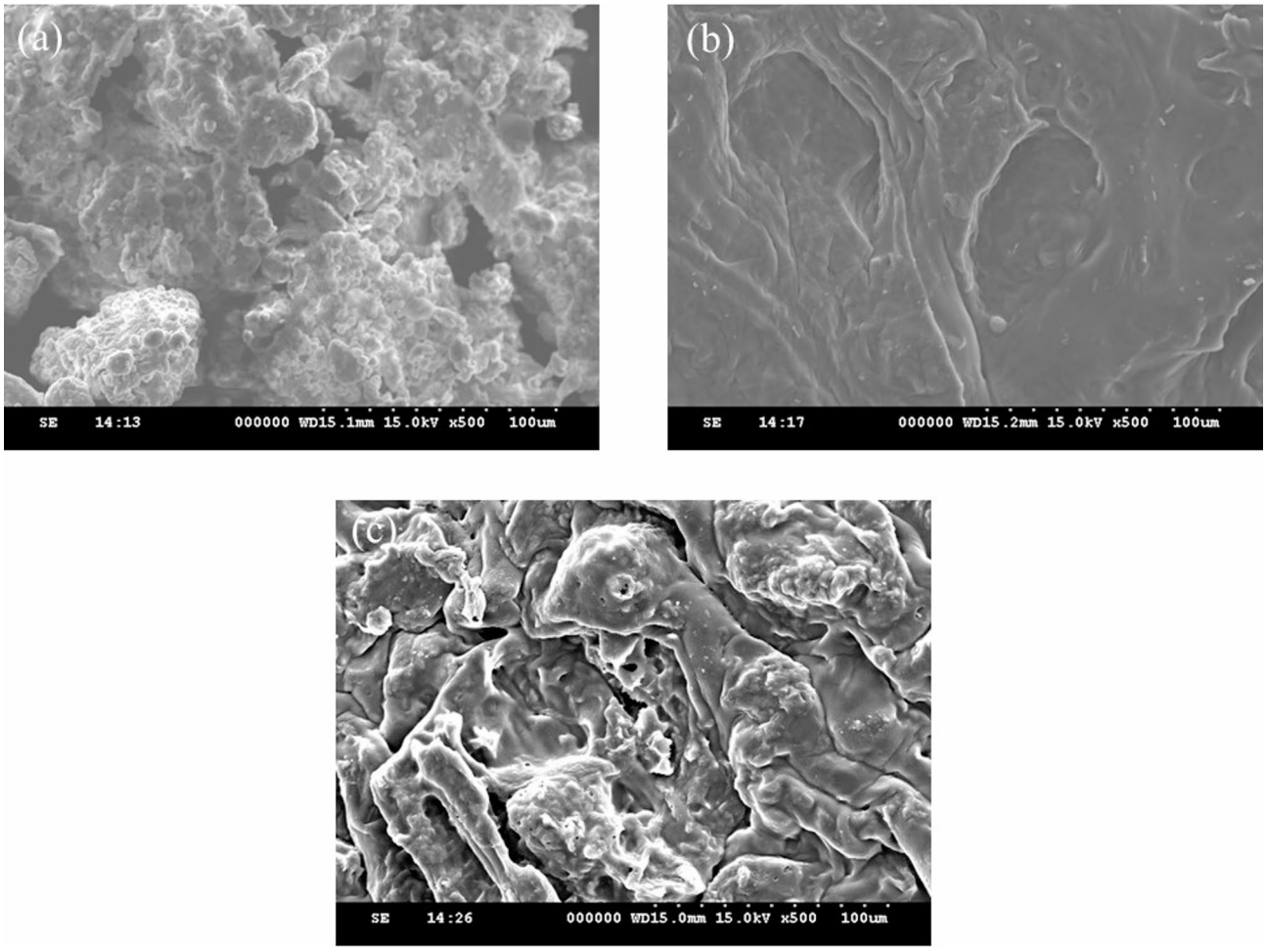

Fig. 2. (**a**) SEM image of RB, (**b**) SEM image of RB / AA, (**c**) SEM image of RASH.

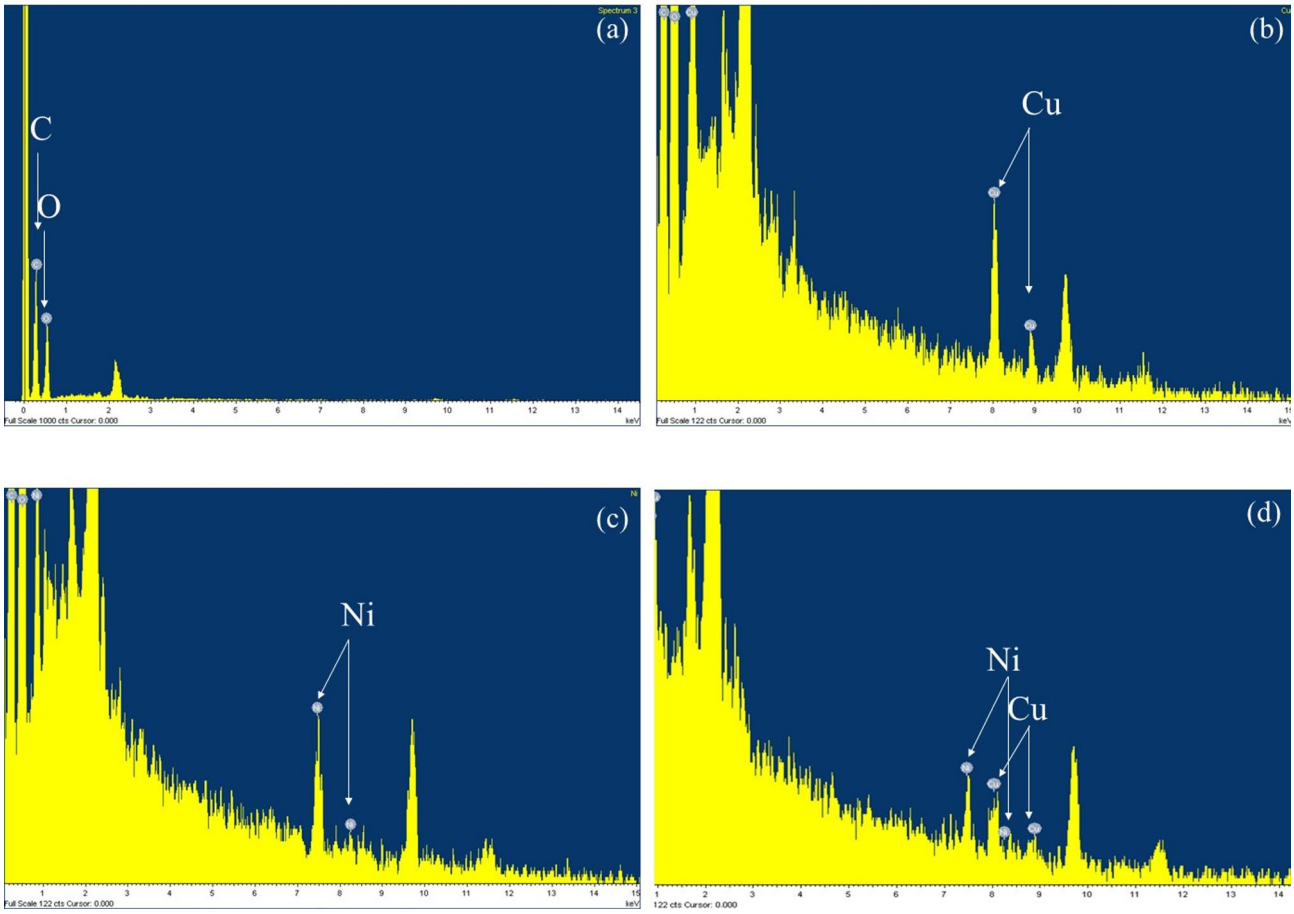

Fig. 3. EDS spectrum of the as-obtained samples. (**a**) Element mapping pattern of RASH, (**b**) Element mapping pattern of RASH (after adsorption of copper ions), (**c**) Element mapping pattern of RASH (after adsorption of nickel ions), (**d**) Element mapping pattern of RASH (after adsorption of copper and nickel ions).

at 1539 cm^{-1} and 1450 cm^{-1} indicated the asymmetric and symmetric vibrations of carboxylic acid[30]. A broad peak observed at 3400–3600 cm^{-1} can be attributed to O-H and N-H stretching vibrations.

In RASH samples, characteristic peaks detected at 1031 cm^{-1} and 1161 cm^{-1} corresponded to C-OH stretching vibrations of the algal salt polysaccharide structure. Peaks at 1701 cm^{-1} and 1449 cm^{-1} corresponded to the asymmetric and symmetrical elongation vibrations of carboxylate (COO^-) groups, while the broad peak at 3466 cm^{-1} corresponded to the tensile vibration of O-H stretching[31,32].

In the FTIR spectra, the intensity and position of the characteristic peaks in RB and RB/PAA differed from those in RASH samples. Moreover, the lower intensity of hydroxyl stretching vibrations in RB and RB/PAA confirmed the anchoring of functional groups. As shown in Fig. 4b, the intensities of all characteristic peaks decreased after adsorption, thereby confirming the interaction of heavy metal ions with the adsorbent.

Figure 5 presents TGA analysis of RB, RB/AA, and RASH samples. We observed distinct differences in the temperature at which the samples began losing weight: RB (76 °C), RB/AA (168 °C), and RASH (174 °C). These findings clearly illustrate the effects of AA and alginate on thermal stability. The initial weight loss (0 and 110 °C) was primarily due to the volatilization of adsorbed water[33]. RB samples exhibited a sharp weight loss between 240 °C and 360 °C, due mainly to the decomposition of cellulose. RB/AA underwent decomposition at 340 °C ~ 500 °C, due mainly to carboxyl group and glutamamine degradation[30], whereas RASH underwent decomposition at 340 °C ~ 500 °C, due to carboxyl group degradation[34].

In a pH-specific environment, the adsorbent surface has zero net charge, referred to as the point of zero charge (pH_pzc)[35]. In this study, we determined the pH_pzc of RASH and analyzed the relationship between pH variation and surface charge to elucidate the interaction between heavy metal ions and the adsorbent. As shown in Fig. 6, when the solution pH exceeded 3.28, the surface of RASH carried a net negative charge due to deprotonation of functional groups such as –OH and –COOH, thereby enhancing electrostatic attraction toward positively charged Cu^{2+} and Ni^{2+} ions. Conversely, at pH values below 3.28, protonation of surface groups led to a net positive charge, resulting in electrostatic repulsion between the adsorbent surface and the cationic metal ions, thus lowering adsorption efficiency[36].

The expansion of the hydrogel adsorbents in an aqueous solution was characterized by comparing the hydrophilicity of RB/AA and RASH hydrogel adsorbents using the weight method. As shown in Fig. 7, water molecules entered the RASH hydrogel network more rapid than the RB/AA during the initial diffusion stage,

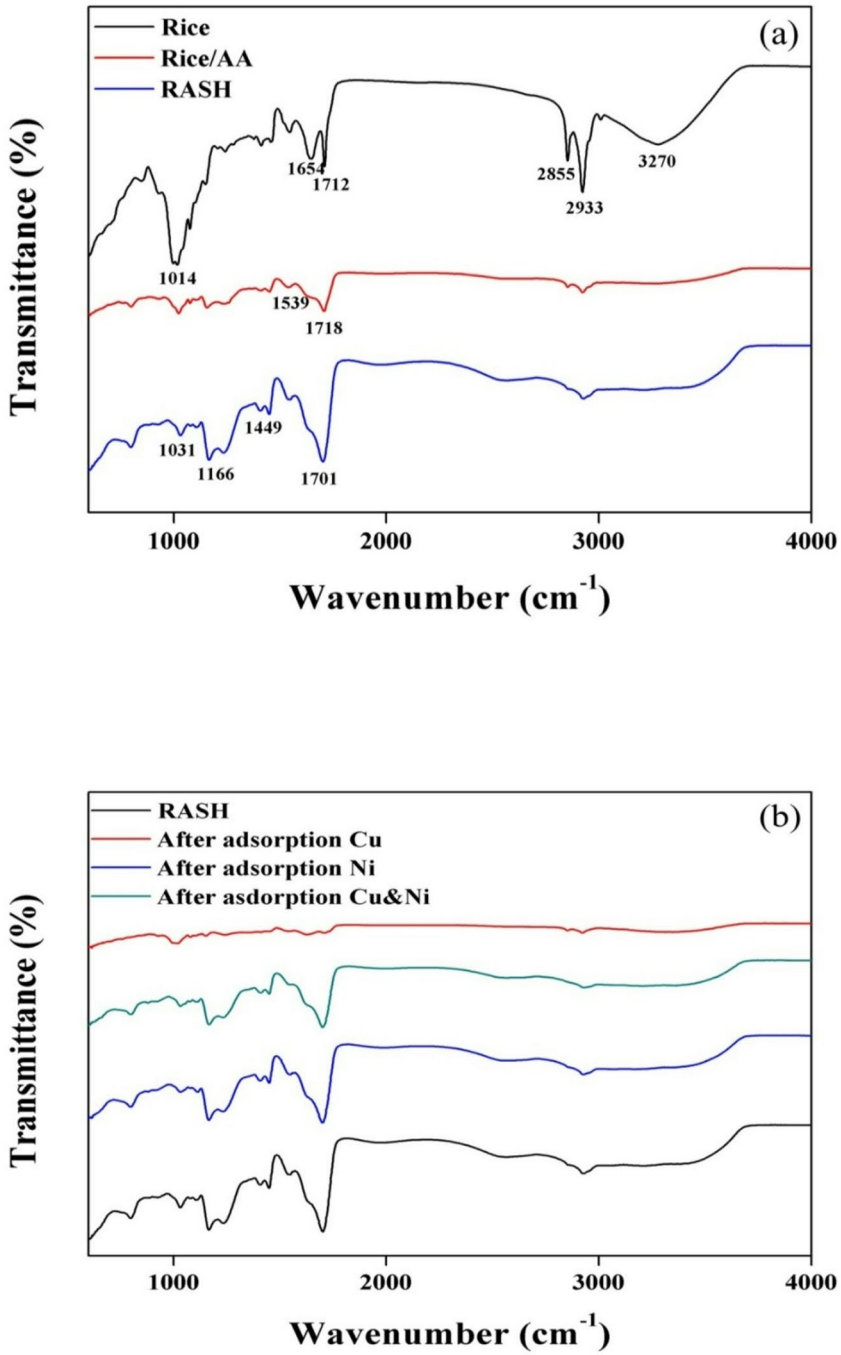

Fig. 4. FTIR spectra of the as-prepared adsorbents.

thereby enhancing the swelling ability of the hydrogel. This effect can be attributed to the electrostatic repulsion produced by the hydrophilic functional groups (carboxyl groups) in the alginate[37]. However, the three-dimensional (3D) interconnected network of channels in RASH samples restricted water penetration, thereby prolonging the time to reach expansion equilibrium[30].

pH is an important factor in the process of copper/nickel ion adsorption. As shown in Fig. 8, the adsorption capacity of the RASH adsorbent increased with an increase in pH. However, the formation of $Cu(OH)_2/Ni(OH)_2$ precipitates under strongly alkaline conditions reduced the overall adsorption capacity[38].

The amount of adsorbent is one of the important factors affecting adsorption efficiency. From Fig. 9, it can be seen that the adsorption capacity for copper and nickel ion reached 13 $mg{\cdot}g^{-1}$ and 10 $mg{\cdot}g^{-1}$, respectively. However, the dose gradually increased to 5 particle, the adsorption capacity for copper/nickel ion gradually decreased. This is because the concentration of the metal ions remained unchanged, the amount of the adsorbent increased, the number of free adsorption sites increased and agglomeration occurs between adsorption sites,

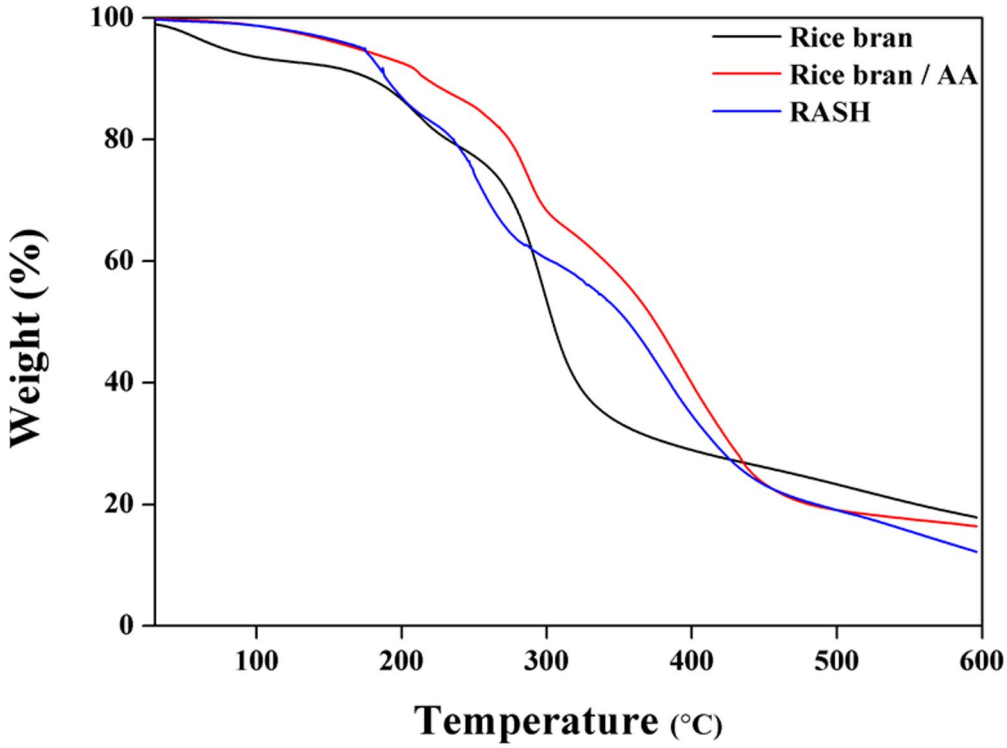

Fig. 5. TGA curves of the as-prepared adsorbents.

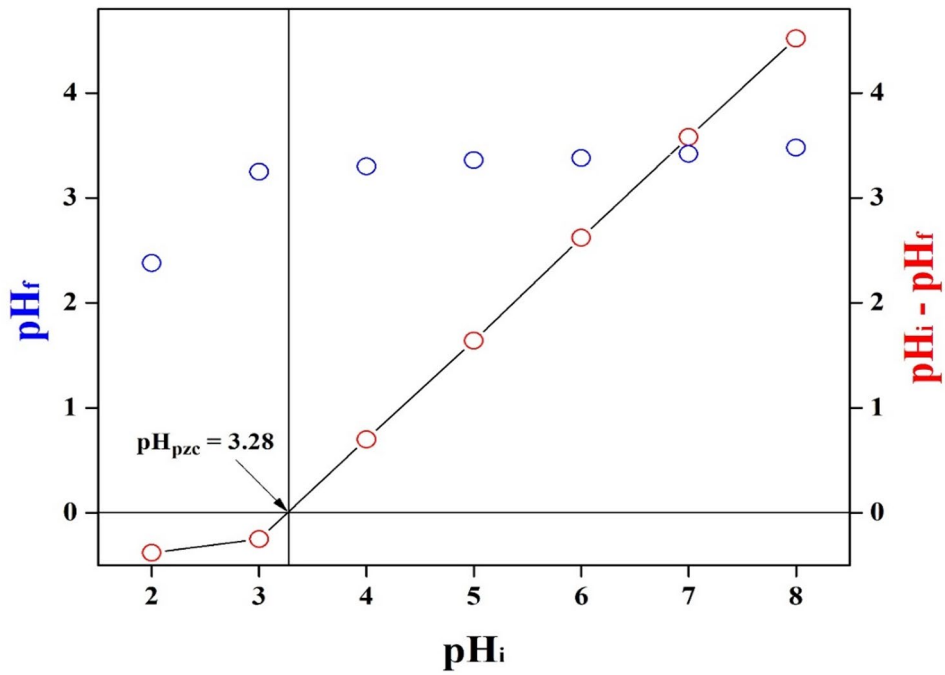

Fig. 6. The zeta potential of RASH under different pH values.

thereby reducing adsorption capacity of the adsorbent[39]. In order to achieve the best removal effect and maximize the efficiency of the adsorbent, 1 particle was selected as the optimal adsorption dose for removing Cu^{2+} / Ni^{2+}.

We also explored the effect of temperature on adsorption capacity using an aqueous solution containing Cu^{2+} ions (100 ppm) at pH 5.5 with an adsorbent dosage of 1 pcs. As shown in Fig. 10 (a), Cu^{2+} adsorption capacity decreased with an increase in temperature from 30 °C to 60 °C, indicating that the adsorption process involves

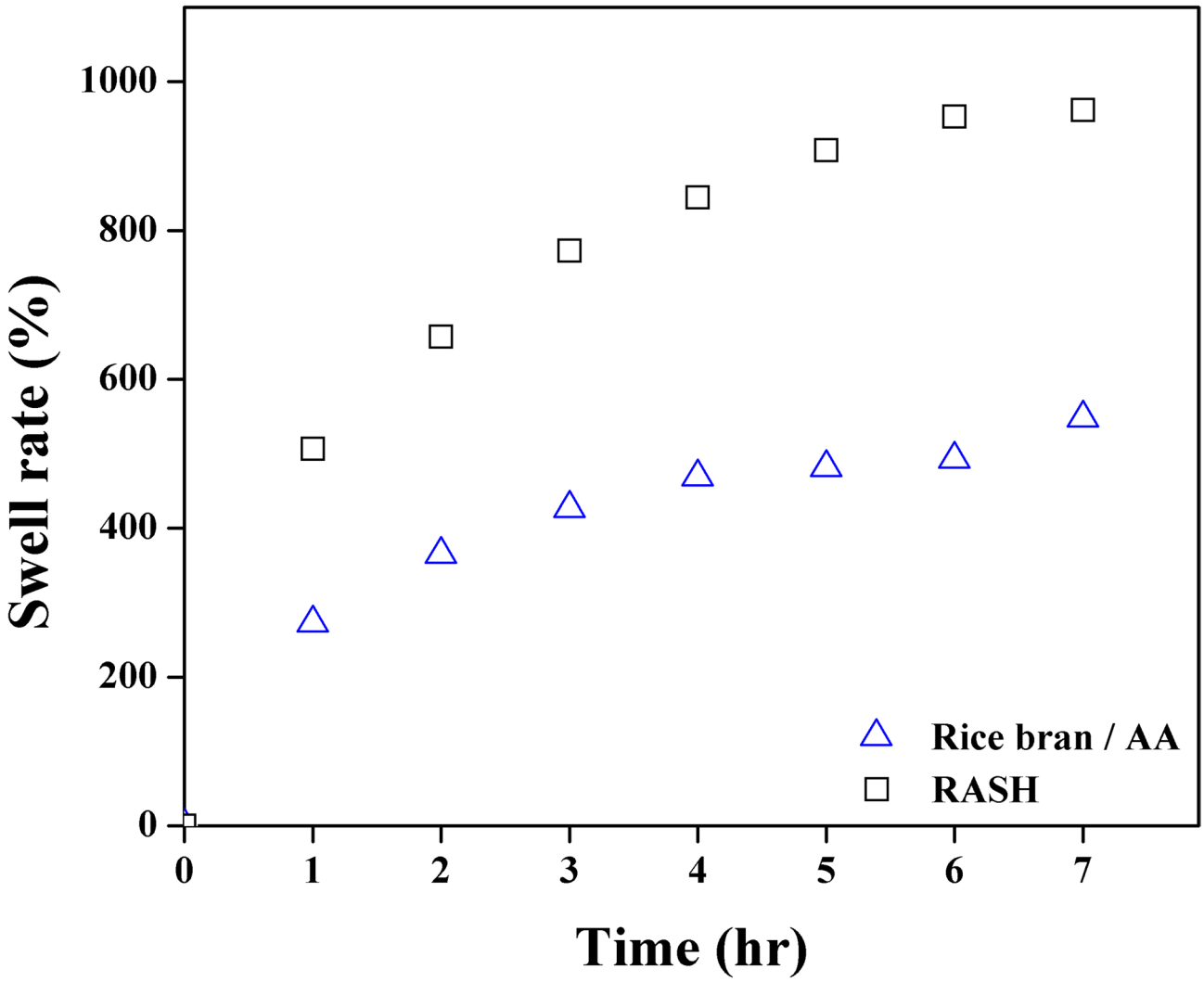

Fig. 7. The expansion characteristics of the as-prepared adsorbents.

an exothermic reaction. The increase in temperature increased the rate of metal ion migration, which led to an increase in the number of metal ions that detached from the adsorbent surface with a corresponding decrease in adsorption[40]. Overall, the optimal temperature for copper ion adsorption was 30 °C.

The same experiment was conducted using an aqueous solution containing Ni^{2+} ions (100 ppm) at pH 8 with adsorbent dosage of 1 pcs. As shown in Fig. 10 (b), the Ni^{2+} adsorption capacity decreased with an increase in temperature, again indicating an exothermic reaction. Again, the optimal adsorption temperature was 30 °C.

To investigate the effect of initial heavy metal concentration on the adsorption capacity, we conducted an experiment using an aqueous solution containing Cu^{2+} or Ni^{2+} ions with an adsorbent dosage of 1 pcs. As shown in Fig. 11 (a), increasing the initial concentration of copper ions from 13 $mg{\cdot}g^{-1}$ to 25 $mg{\cdot}g^{-1}$ led to a gradual increase in adsorption capacity; however, adsorption performance decreased beyond this ion concentration. This can be explained by the fact that the RASH adsorbent possessed sufficient functional groups (carboxyl groups) to react a moderate number of Cu^{2+} ions, but not enough sites to accommodate high concentrations (adsorption saturation). As shown in Fig. 11 (b), increasing the initial concentration of Ni ions from 10 $mg{\cdot}g^{-1}$ to 28 $mg{\cdot}g^{-1}$ had the same effect as the Cu.

Isothermal adsorption models describe the interaction between the adsorbate and adsorbent in the solution. In the current study, the isothermal adsorption of RASH adsorbent was assessed from the perspectives of the Langmuir, Freundich, and Dubin Radushkevich (D-R) isothermal equations.

As shown in Table 1, the process of Cu^{2+} adsorption on RASH conformed to the Langmuir isotherm model ($R^2 > 0.99$), indicating mono-layer adsorption with a maximum adsorption capacity (qm) of 26.50 $mg{\cdot}g^{-1}$. The separation factor ($R_{L)}$ ranged from 0.1783 to 0.2569, confirming a favorable adsorption process. A Freundlich isotherm parameter (n^{-1}) ranging from 0.18 to 0.21 ($0 < R_L < 1$) confirmed a favorable adsorption process[35].

The Dunin-Radushkevich (D–R) model yielded an average free energy (Ea) ranging from 0.15 to 0.17 $kJ{\cdot}mol^{-1}$ (less than 8 $kJ{\cdot}mol^{-1}$), which suggests that physical adsorption was the dominant process[41].

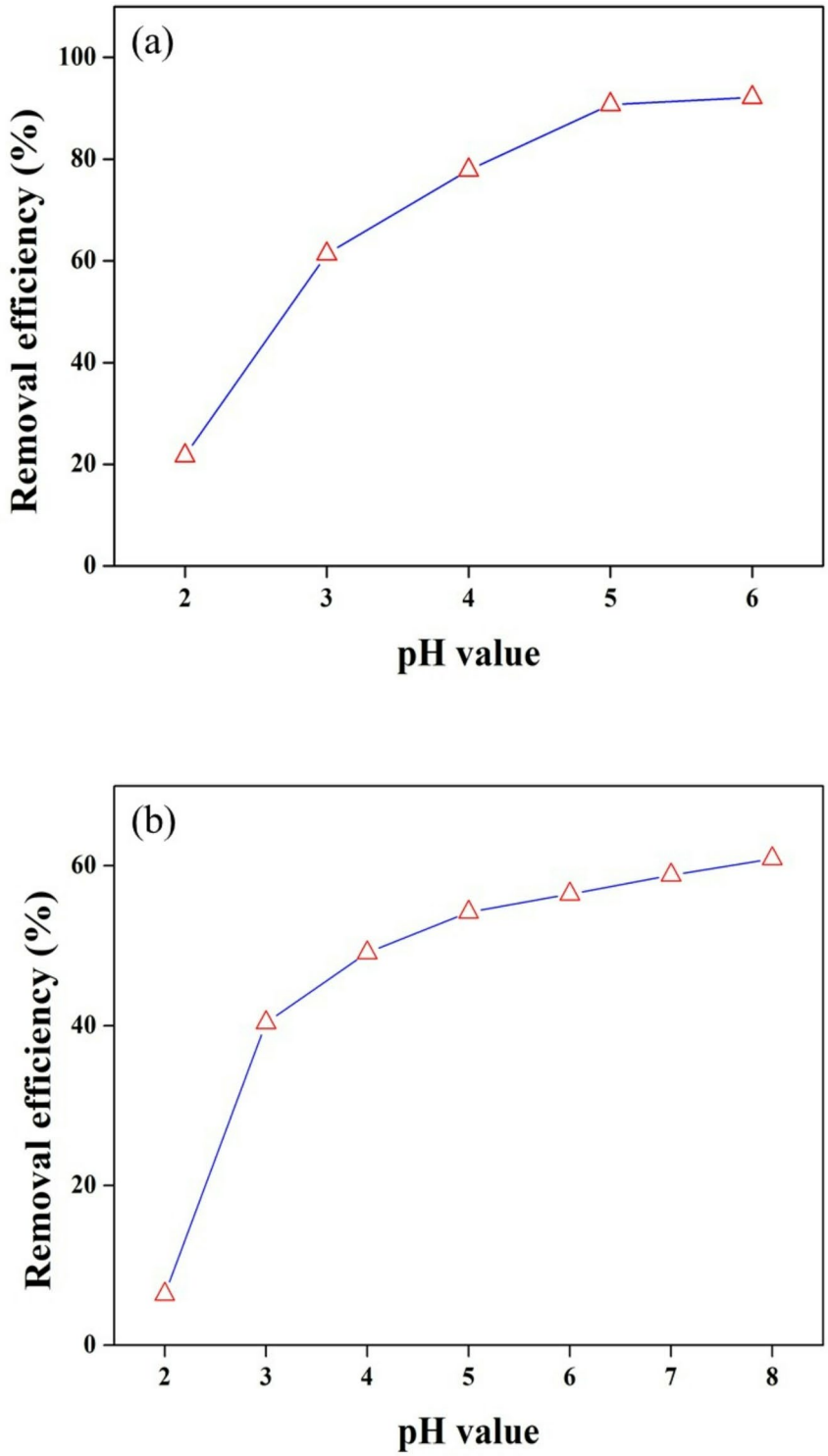

Fig. 8. The influence of pH value on the adsorption performance for metal ions removal (**a**) Cu^{2+}, (**b**) Ni^{2+}.

As shown in Table 2, the process of Ni ion adsorption on RASH also conformed to the Langmuir isotherm adsorption model, indicating mono-layer adsorption with a maximum adsorption capacity (q_m) of 41.81 mg·g^{-1}. The R_L value ranged was 0.4005 to 0.6068 and n^{-1} ranged from 0.39 to 0.49, indicating a favorable adsorption process. The D–R model yielded an average E_a range of 0.024 to 0.04 kJ·mol^{-1}, confirming that physical adsorption governed the process.

As shown in Fig. 12, the adsorption process can be divided into three stages[42]. In the first stage, the adsorbent had a large number of active sites driving rapid adsorption. In the second stage, the diffusion of metal ions

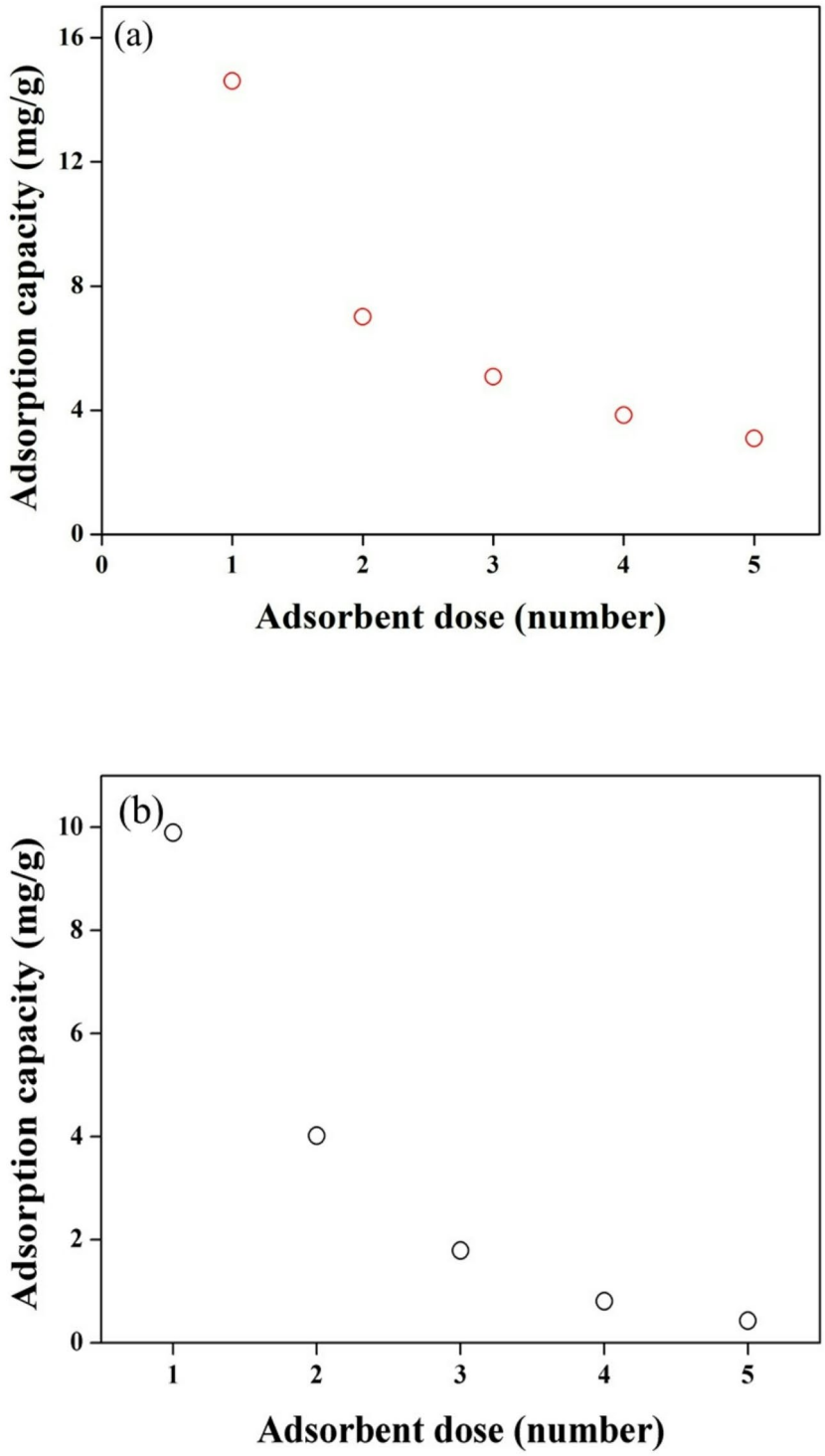

Fig. 9. The influence of adsorbent dose on the adsorption performance for metal ions removal (**a**) Cu^{2+}, (**b**) Ni^{2+}.

into the pores of the adsorbent slowed the rate of adsorption. In the third stage, adsorption reached a dynamic equilibrium.

We sought to describe the dynamic adsorption behavior of the RASH adsorbent from the perspective of pseudo-first-order, pseudo-second-order, and intraparticle diffusion models.

As shown in Table 3; Fig. 13, the pseudo-second-order kinetic model best described the adsorption kinetics of Cu^{2+} on RASH adsorbent, providing a better fit that the pseudo-first-order model. The calculated q_e (9.34 m $g{\cdot}g^{-1}$) closely aligned with the q_e obtained in experiments (8.31 $mg{\cdot}g^{-1}$), confirming that the adsorption process followed the pseudo-second-order kinetic model.

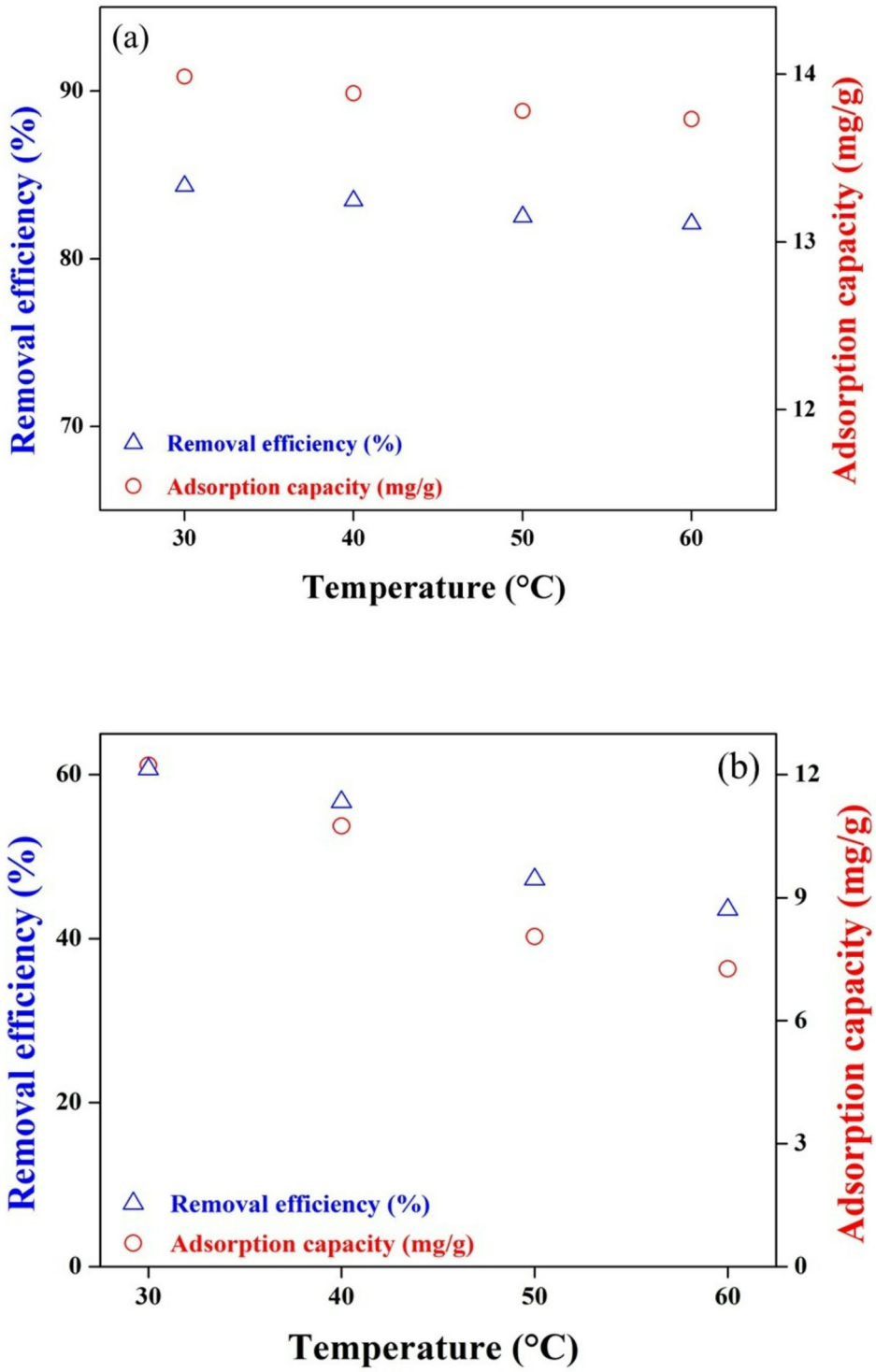

Fig. 10. The influence of temperature on the adsorption performance for metal ions removal (**a**) Cu^{2+}, (**b**) Ni^{2+}.

As shown in Table 4; Fig. 14, the pseudo-second-order kinetic model best described the adsorption kinetics of Ni^{2+} on RASH. Again, the calculated q_{exp} (9.34 mg·g^{-1}) closely aligned with the value obtained in experiments (8.29 mg·g^{-1}), confirming that the adsorption process followed the pseudo-second-order kinetic model.

The intraparticle diffusion model was used to elucidate the mechanism underlying the adsorption of Cu and Ni ions on RASH. As shown in Fig. 15, the adsorption process exhibited a multi-linear profile, indicative of multiple adsorption stages. The fact that intercept (C) was not zero further indicates that the internal diffusion of particles was not the only rate-controlling mechanism involved in the adsorption process.

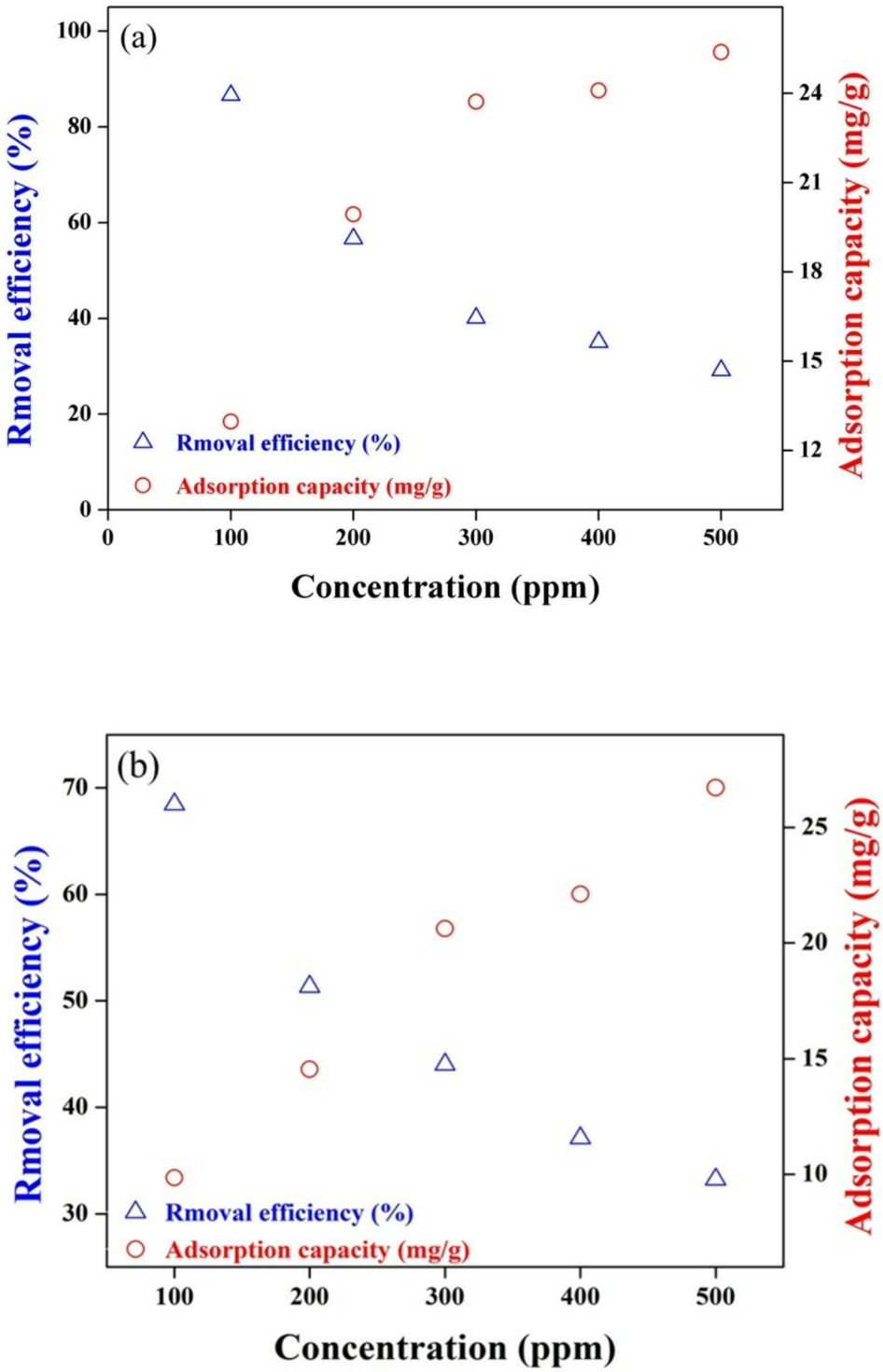

Fig. 11. The influence of initial concentration on the adsorption performance for metal ions removal (**a**) Cu^{2+}, (**b**) Ni^{2+}.

As shown in Tables 3 and 4, the boundary layer effect increased with an increase in intercept values (C), hindering ion diffusion to active sites. The diffusion rate constants followed the trend $k_id_1 > k_id_2$, indicating that in the initial stage of diffusion, the adsorption of Cu^{2+} and Ni^{2+} ions on the RASH can be attributed to the abundance of active sites[43]. Comparison of heavy metal adsorption in wastewater by agricultural waste, as shown in Table 5[44–46].

Ion	Models	Parameters	T(℃)			
			30	40	50	60
Cu^{2+}	Langmuir	q_m (mg/g)	26.50	26.48	26.06	25.78
		K_L	0.0469	0.0346	0.0295	0.0289
		R^2	0.9971	0.9933	0.9663	0.9651
		R_L	0.1783	0.2339	0.2525	0.2569
	Freundlich	n	4.84	4.843	5.82	5.56
		K_F	7.74	7.42	8.59	8.13
		R^2	0.9856	0.9855	0.9007	0.9038
	Duinin-Radushkevich	q_m (mg/g)	23.33	22.21	21.45	21.54
		E_a (kJ/mol)	0.17	0.16	0.16	0.15
		R^2	0.8716	0.7086	0.6176	0.6743

Table 1. Isothermal adsorption model parameters for Cu^{2+} adsorption.

Ion	Models	Parameters	T(℃)			
			30	40	50	60
Ni^{2+}	Langmuir	q_m (mg/g)	41.81	31.45	29.63	25.49
		K_L	0.01305	0.0088	0.0071	0.0067
		R^2	0.9565	0.9028	0.9156	0.9883
		R_L	0.4005	0.4972	0.5946	0.6068
	Freundlich	n	2.59	2.7	2.15	2.06
		K_F	3.78	2.7	1.41	1.06
		R^2	0.9440	0.8887	0.9241	0.9888
	Duinin-Radushkevich	q_m (mg/g)	28.16	20.19	17.89	16.06
		E_a (kJ/mol)	0.075	0.055	0.049	0.04
		R^2	0.5663	0.4647	0.5718	0.7770

Table 2. Isothermal adsorption model parameters for Ni^{2+} adsorption.

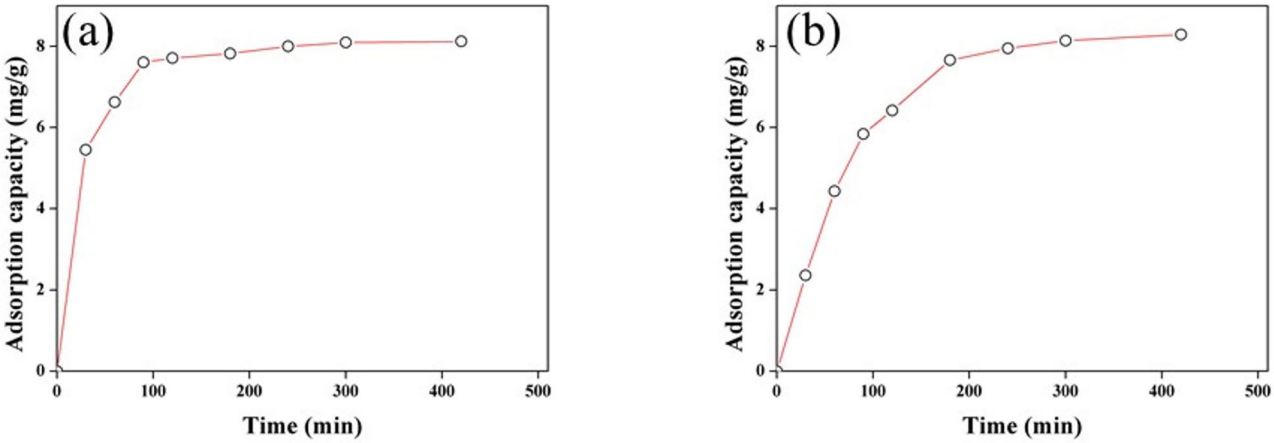

Fig. 12. Dynamic adsorption curve of metal ions by RASH (**a**) Cu^{2+}, (**b**) Ni^{2+}.

The recycling ability of an adsorbent is crucial to the cost-effectiveness of water treatment. This study explored the reusability of RASH by conducting cyclic experiments. As shown in Fig. 16, the removal rate of Cu^{2+} was 80% after 5 adsorption-desorption cycles, while the removal rate of Ni^{2+} was 58%, demonstrating good reusability.

Conclusions

A novel hydrogel adsorbent (RASH) was successfully synthesized from RB, AA, and SA for the efficient removal of Cu^{2+} and Ni^{2+} from aqueous solutions. The hydrogel exhibited maximum adsorption capacities of 26.5 mg g^{-1} for Cu^{2+} (88% removal efficiency) and 41.81 mg g^{-1} for Ni^{2+} (62% removal efficiency) under optimal conditions.

Ion	Model	Kinetic parameters	
Cu^{2+}	Experiment	qexp(mg/g)	8.29
	Pseudo-first-order	k_1(1/min)	0.0084
		q_e (mg/g)	2.8
		R^2	0.7739
	Pseudo-second-order	k_2 (g/mg·min)	0.0126
		q_e (mg/g)	8.31
		h (mg/g·min)	0.87
		R^2	0.9987
	External diffusion	K_{id1}	0.8145
		C_1	0.2965
		R^2	0.9679
	Equilibrium diffusion	K_{id2}	0.0502
		C_2	7.164
		R^2	0.9269

Table 3. Dynamic adsorption model parameters for Cu^{2+} adsorption.

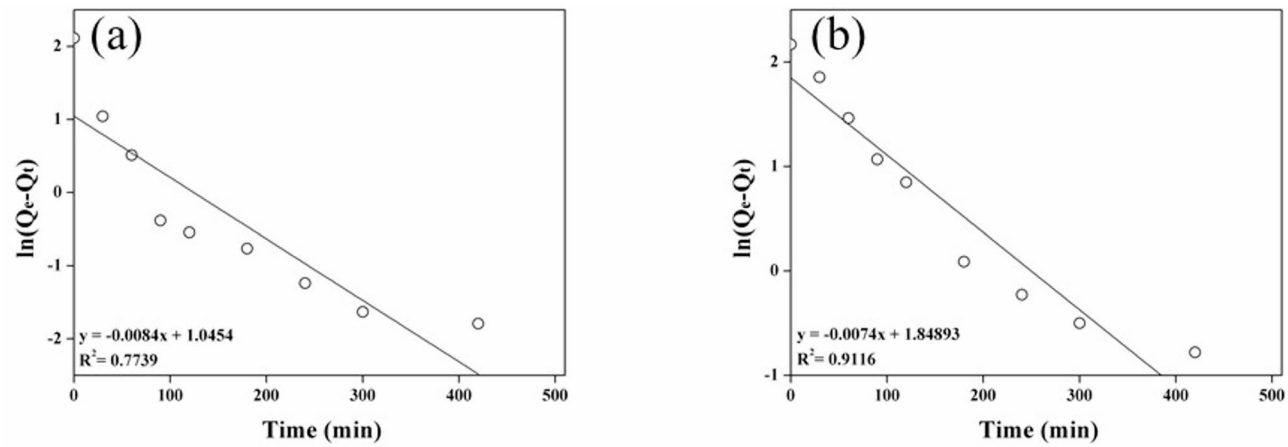

Fig. 13. Pseudo-first-order model curve (**a**) Cu^{2+}, (**b**) Ni^{2+}.

Ion	Model	Kinetic parameters	
Ni^{2+}	Experiment	qexp(mg/g)	8.75
	Pseudo-first-order	k_1(1/min)	0.0074
		q_e (mg/g)	6.35
		R^2	0.9116
	Pseudo-second-order	k_2 (g/mg·min)	0.00218
		q_e (mg/g)	9.34
		h (mg/g·min)	0.19
		R^2	0.9671
	External diffusion	K_{id1}	0.5981
		C_1	-0.2423
		R^2	0.9778
	Equilibrium diffusion	K_{id2}	0.0654
		C_2	6.9676
		R^2	0.8982

Table 4. Dynamic adsorption model parameters for Ni^{2+} adsorption.

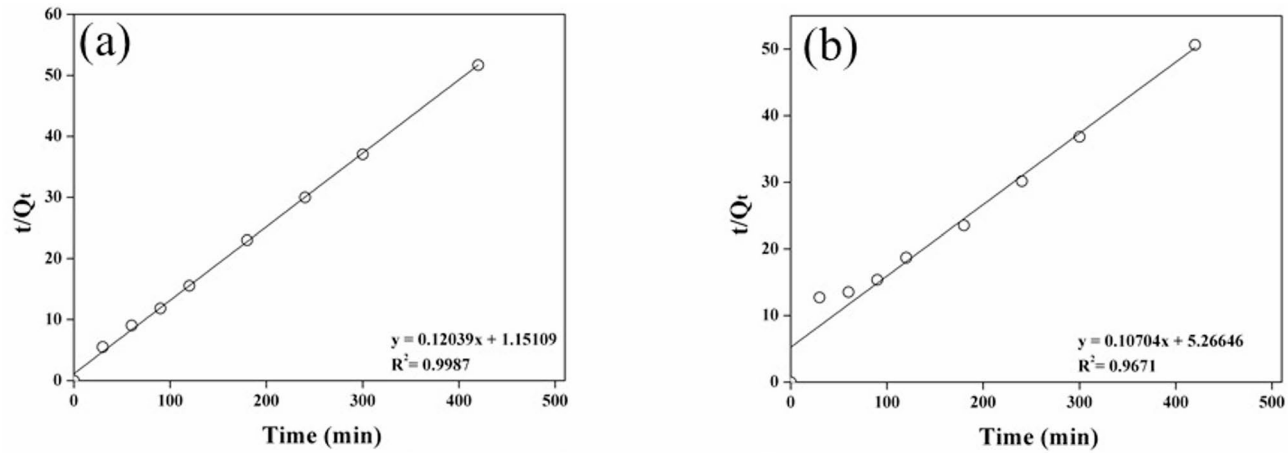

Fig. 14. Pseudo-second-order model curve (**a**) Cu^{2+}, (**b**) Ni^{2+}.

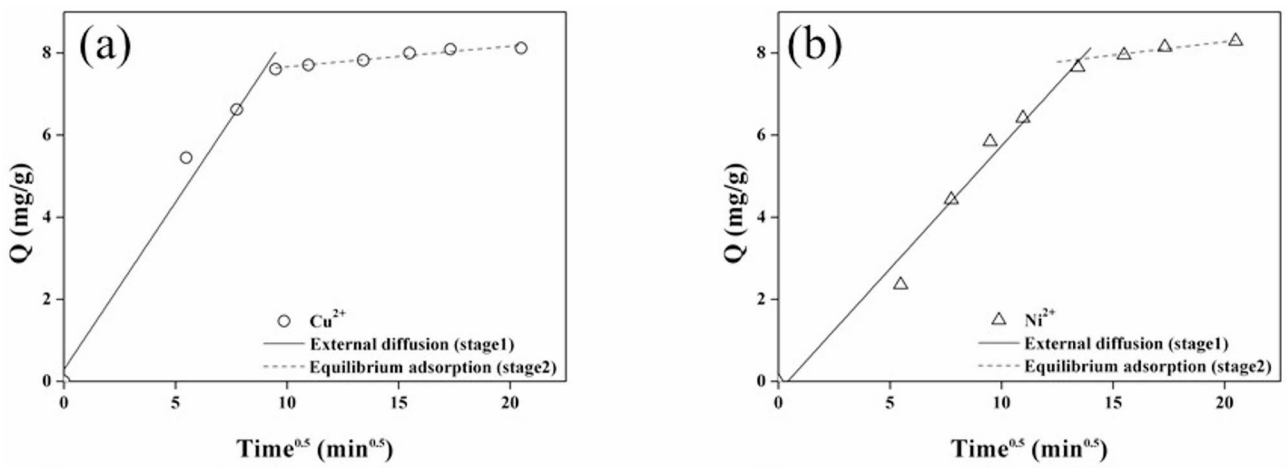

Fig. 15. Internal diffusion model curve (**a**) Cu^{2+}, (**b**) Ni^{2+}.

Adsorbent	Pollutants	Adsorption Capacity (mg/g)	References
Kaolinite	Cu(II)	4.42	44
Watermelon shell	Cu(II)	111.10	45
Bamboo Shoot Shell	Re (VII)	24.5	42
Sawdust Carbon	Cd(II)	63.3	46
Rice bran	Cu(II) Ni(II)	26.5 41.18	This is work

Table 5. Comparison of heavy metal adsorption in wastewater by agricultural waste.

Adsorption behavior conformed to the Langmuir isotherm model, indicating monolayer adsorption, while Dubinin–Radushkevich analysis suggested physical adsorption as the dominant mechanism. The hydrogel maintained over 85% of its initial adsorption capacity after five reuse cycles, demonstrating strong regeneration ability. Given its low production cost, environmental friendliness, and robust adsorption performance, the RASH hydrogel shows significant potential for scalable industrial wastewater treatment. Future research could explore the use of other agricultural wastes (such as fruit shells, stems, and peels) as materials for hydrogel synthesis to achieve excellent and accessible low-cost and sustainable adsorbents.

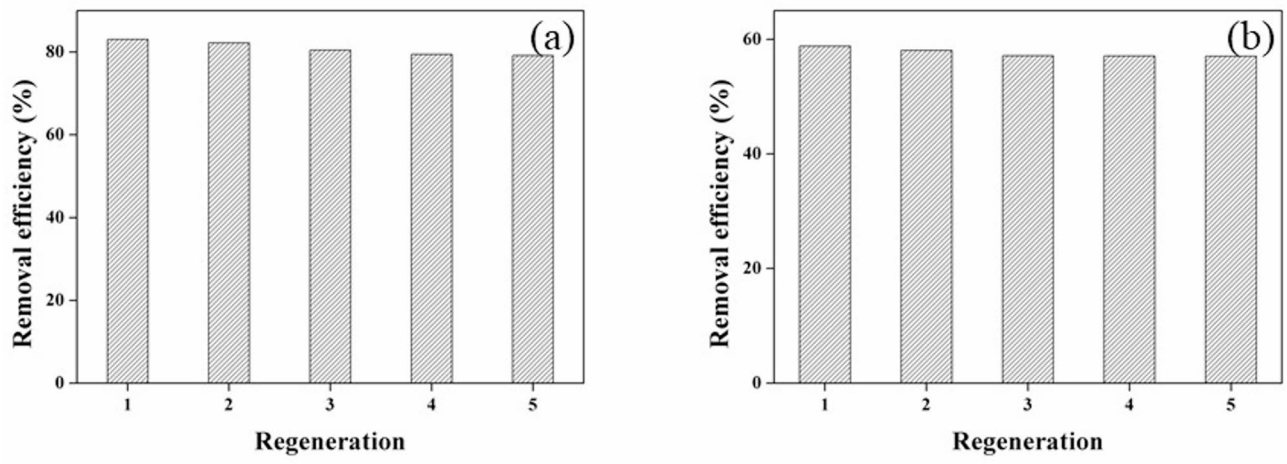

Fig. 16. Recycling stability tests over as-obtained RASH.

Data availability

All data generated and analyzed during this study are included in this published article. Further information and additional datasets are available from the corresponding author upon reasonable request.

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Author contributions
Gui-Bing Hong: Conceptualization, writing original draft, investigation, visualization. Yue-Hui Lin: Conceptualization. Wang-De Lin: Conceptualization, supervision, resources, data analysis, methodology.

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Declarations

Competing interests
The authors declare no competing interests.

Additional information
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